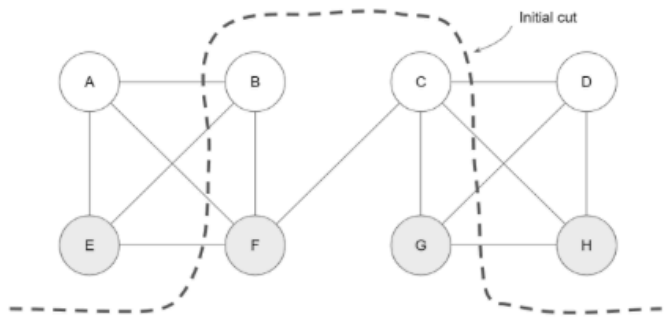
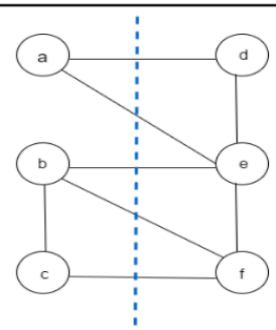
Previous Years Question

Physical Design & KL Algorithm

Q-1:

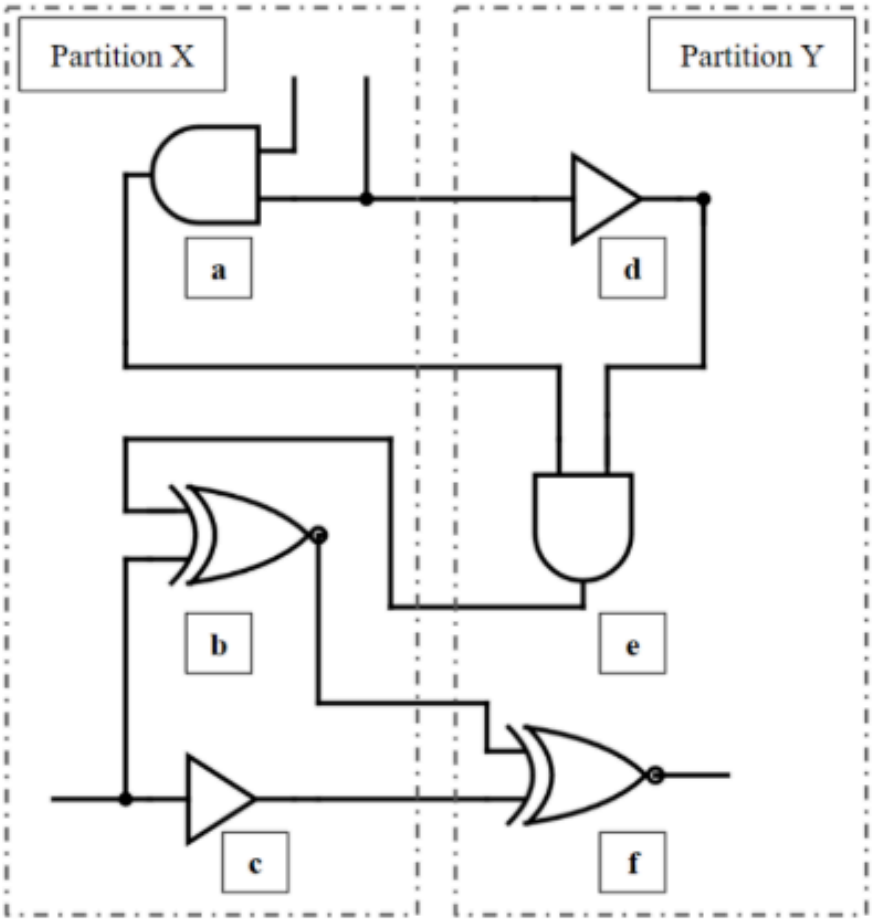
For a certain chip planning task, you are given three blocks with the following widths and heights: Block A: $w_A = 1$, $h_A = 4$ or $w_A = 4$, $h_A = 1$ or $w_A = 2$, $h_A = 2$ Block B: $w_B = 1$, $h_B = 2$ or $w_B = 2$, $h_B = 1$ Block C: $w_C = 1$, $h_C = 3$ or $w_C = 3$, $h_C = 1$		
(a)	**Define** chip planning. **Find** and **draw** a floorplan with the *minimum* total area enclosed by its global bounding box.	6
An initial cut for a partition of 8 circuit elements (A, B, C, D, E, F, G, and H) is given below:  		
(b)	**Calculate** the *initial* cut cost, and the cost of moving each node.	6
(c)	By inspection (without using an algorithm or any calculation), propose a better cut for the circuit.	3

Q-2:

An initial cut for a partition of 6 circuit elements (**a, b, c, d, e,** and **f**) is given in **Figure 1**. The cut line divides the circuit elements into two partitions.			
(a)	**Identify** how many iterations are required in a single pass.	**[1]**	
(b)	**Calculate** the initial cut cost and cost of moving each node.		**[4]**
(c)	**Perform** the first pass of the algorithm. For every "**i**"th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: i. Compute the node costs of all unfixed nodes ii. Find the maximum gain of swapping a pair of nodes (Δg_i) iii. Swap the pair and draw the updated graph Finally, find the maximum G_m and redraw the optimized graph.		**[9]**
(d)	Should you perform subsequent passes of the algorithm? Why or why not?		**[1]**

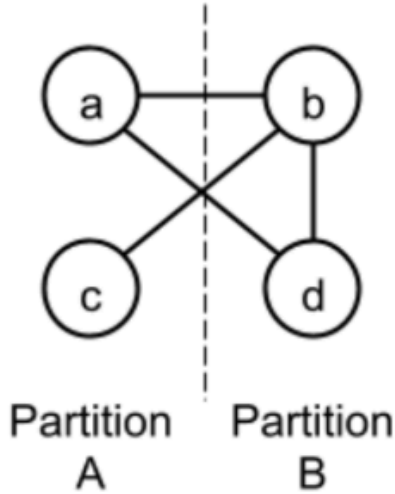
Q-3:

Consider the following circuit with 6 logic gates (a-f) and the initial partitioning marked by the dotted lines.



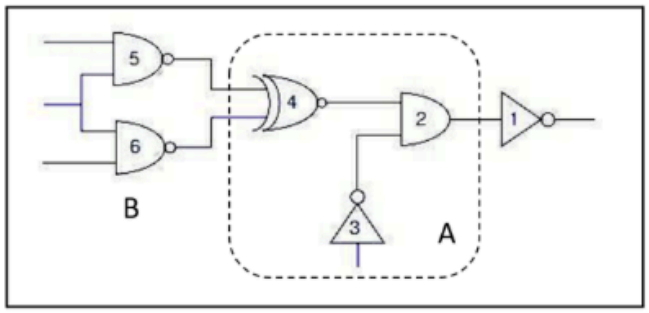
(a)	**Represent** the given circuit and the initial partitioning using a graph.	**[3]**
(b)	**Compute** the initial cut cost.	**[1]**
(c)	**Perform** the first pass of the algorithm. For every "i"th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: i. Compute the node costs of all unfixed nodes ii. Find the maximum gain of swapping a pair of nodes (Δg_i) iii. Swap the pair and draw the updated graph	**[6]**
(d)	Finish the first pass by computing the maximum positive gain, G_m. **Suggest** how many swaps should be actually executed in the first pass.	**[4]**
(e)	Should you perform subsequent passes of the algorithm? Why or why not?	**[1]**

Q-4:

The graph below (nodes a-d) can be optimally partitioned using the Kernighan-Lin algorithm. The dotted line represents the initial partitioning. Assume all the edges have the same weight.  	
i. **Calculate** the initial cut cost.	**1**
ii. **Identify** how many iterations are needed in a single pass?	**1**
iii. **Perform** the first pass of the algorithm and **determine** the new cut cost of the optimum output of the first pass. [Hint: For the "i"th iteration of the first pass, until all the nodes are swapped and fixed, do the following: • Compute the node costs of all unfixed nodes • Find the maximum gain of swapping a pair of nodes (Δg_i) • Swap the pair and draw the updated graph • Calculate cumulative gain, G_i]	**7**
iv. Should you perform subsequent passes of the algorithm? Why or why not?	**1**

Q-5:

Mr. Xin Chao is a VLSI engineer in a renowned chip manufacturing company. He sets the physical positions of all the components on a chip to optimize the delay and power consumption. Suppose, Mr. Xin is working on a chip that consists of 6 nodes illustrated in the figure below. Firstly, he partitions the nodes into two equal-sized blocks, and then, using the Kernighan-Lin algorithm, he minimizes the number of connections between the blocks. After floor-planning, placement, and maintaining all other design considerations, Mr. Xin can dispatch the chip to the fabrication unit.



The dotted line in the figure represents the initial partitioning with **A (2,3,4)** & **B (1,5,6)** blocks. <u>For the *first* iteration</u>, answer the following:

(a)	**Draw** the corresponding graph representation of the above circuit.	**2**
(b)	**Find** the initial cut cost.	**2**
(c)	**Calculate** the cost of each node.	**3**
(d)	**Evaluate** necessary gains (Δg) for the first iteration.	**4**
(e)	**Perform** the first swap and **compare** the new cut cost with the initial one.	**4**

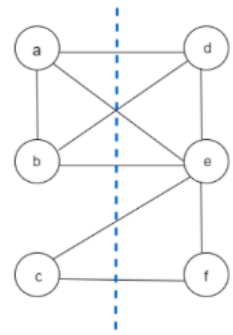
Q-6:

For question (a) you are given a certain chip planning task. You have three blocks with the following widths and heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 3, h = 3 or w = 1, h = 9
For questions (b-e) refer to the adjacent figures at the bottom of the questionr. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown.

(a)	**Find** and draw a floor plan with the minimum total area enclosed by the global bounding box using 10x10 space.	**[3]**

Q-7:

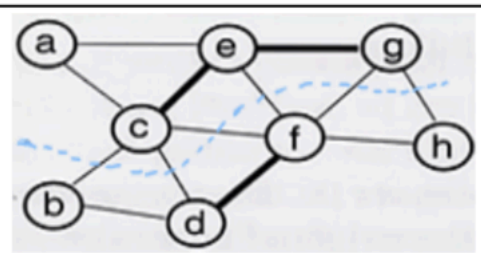
An initial cut for a partition of 6 circuit elements (**a, b, c, d, e,** and **f**) is given in **Figure 1**.
The cut line divides the circuit elements into two partitions.



(a)	**Identify** how many iterations are required in a single pass.	**[1]**
(b)	**Calculate** the initial cut cost and cost of moving each node.	**[4]**
(c)	**Perform** the first pass of the algorithm. For every "**i**"th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: i. Compute the node costs of all unfixed nodes ii. Find the maximum gain of swapping a pair of nodes (Δg_i) iii. Swap the pair and draw the updated graph Finally, find the maximum G_m and redraw the optimized graph.	**[9]**
(d)	Should you perform subsequent passes of the algorithm? Why or why not?	**[1]**

Q-8:

Figure 1 shows an initial partitioning of 8 nodes (**a, b, c, d, e, f, g, h**).
The cut line divides the circuit elements into two partitions.



(a)	**What** is the initial cut cost? **Compute** the costs of moving each node from one partition to another.	**[1+1]**
(b)	**Perform** the first iterations of the first pass of the KL algorithm i. Find the maximum gain of swapping a pair of nodes (Δg_1) in the first iteration. ii. Swap the pair and draw the updated partition after the first iteration.	**[6]**
(c)	Now **perform** the second iteration of the first pass of the algorithm on the updated partition you got in 1(b) and find the maximum gain (Δg_2) for the second iteration.	**[4]**
(d)	If the maximum gain in the next 2 iterations is $\Delta g_3 = -\Delta g_1$ and $\Delta g_4 = -\Delta g_2$, then after how many swaps do we get the optimized partition? What will be the maximum cumulative gain (Gm) and the new cut cost?	**[3]**

Q-9:

An initial cut for a partition of 8 circuit elements (**a, b, c, d, e, f, g**, and **h**) is given in **Figure 1**.

The cut line divides the circuit elements into two partitions named **A** and **B**.

(a) **Calculate** the initial cut cost, and the cost of moving each node. **[1]**

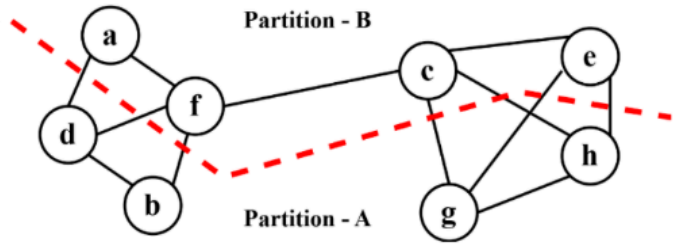


Figure 1

(b) **Perform** the first pass of the algorithm. For every "**i**"th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: **[9]**

i. Compute the node costs of all unfixed nodes
ii. Find the maximum gain of swapping a pair of nodes (Δg_i)
iii. Swap the pair and draw the updated graph

(c) **Compute** G_i of each iteration and find the maximum $G_i = G_m$. **Suggest** how many swaps should be actually executed in the first pass. Redraw the optimized graph. **[4]**

(d) Should you perform subsequent passes of the algorithm? Why or why not? **[1]**

Q-10:

For question (a) you are given a certain **chip planning task.** You have three blocks with the following widths and heights:

Block A: $w = 2, h = 3$ or $w = 3, h = 2$
Block B: $w = 1, h = 2$ or $w = 2, h = 1$
Block C: $w = 1, h = 4$ or $w = 4, h = 1$

(a) **Define** chip planning. **Find** and draw a floorplan with the minimum total area enclosed by its global bounding box. **[3]**

Q-11:

For question (a), you are given a certain chip planning task. You have four blocks with the following widths and
heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 9, h = 1 or w = 3, h = 3